Python Notes

Sample Study Material for RAG Testing

Introduction to Python

Python is a high-level, interpreted programming language known for its simple and readable syntax. It is widely used in web development, data science, artificial intelligence, automation, and software development.

Variables and Data Types

Variables store values in memory. Common Python data types include int for integers, float for decimal numbers, str for text, bool for True or False values, list for ordered collections, tuple for fixed collections, set for unique values, and dict for key-value pairs.

Conditional Statements

Python uses if, elif, and else statements to make decisions. A condition is evaluated as True or False, and the corresponding block of code is executed.

Loops

A for loop is commonly used to iterate over sequences such as lists and strings. A while loop repeats while a condition remains True.

Functions

A function is a reusable block of code defined using the def keyword. Functions can accept parameters and can return values using the return statement.

Lists and Dictionaries

Lists are ordered and mutable collections. Dictionaries store information as key-value pairs. Both are commonly used for storing and processing data.

Object-Oriented Programming

Python supports classes and objects. A class is a blueprint, while an object is an instance of a class. Important concepts include encapsulation, inheritance, polymorphism, and abstraction.

Example

Example: def add(a, b): return a + b. Calling add(2, 3) returns 5.